

Comparative Analysis — Solutions

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COURSE : COMPUTER VISION

1. Object Detection Performance Comparison

Model A: TP = 80, FP = 20, FN = 30

Model B: TP = 90, FP = 15, FN = 20

(a) Precision and Recall

Precision = $TP / (TP + FP)$ Recall = $TP / (TP + FN)$

Model	Precision	Recall
Model A	$80 / (80+20) = 0.80$ (80%)	$80 / (80+30) = 0.727$ (72.7%)
Model B	$90 / (90+15) = 0.857$ (85.7%)	$90 / (90+20) = 0.818$ (81.8%)

(b) Comparison and Justification

Model B outperforms Model A on both metrics: its precision (85.7%) is higher than Model A's (80%), meaning Model B produces fewer false alarms relative to its detections. Model B's recall (81.8%) is also higher than Model A's (72.7%), meaning Model B misses fewer actual objects. Since Model B has both higher precision and higher recall, it provides better overall object detection performance than Model A.

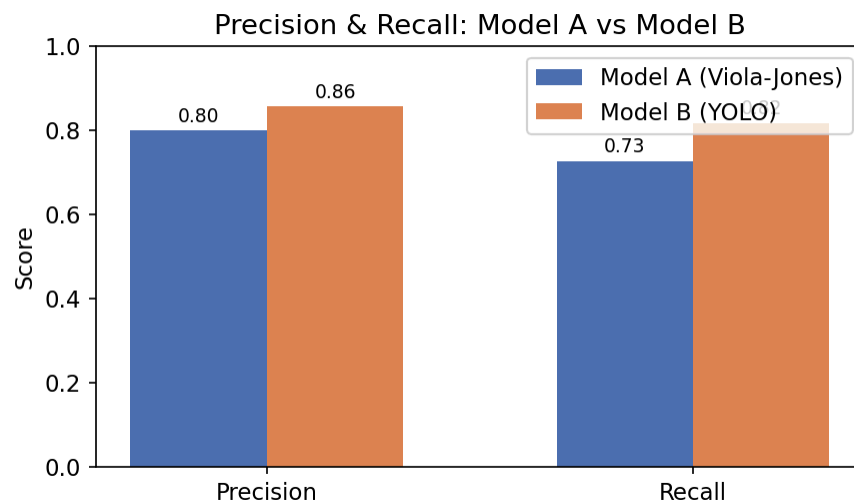


Figure 1: Precision and Recall comparison between Model A (Viola-Jones) and Model B (YOLO).

2. Motion Estimation Error Analysis

Method A displacement errors (pixels): 2, 3, 2

Method B displacement errors (pixels): 3, 5, 4

(a) Average Displacement Error

Method	Errors (px)	Average Error
Method A	2, 3, 2	$(2+3+2)/3 = 2.33$ px
Method B	3, 5, 4	$(3+5+4)/3 = 4.00$ px

(b) Comparison

Method A has a lower average displacement error (2.33 px) compared to Method B (4.00 px).

(c) Justification

A lower average displacement error means the estimated motion is closer to the true motion between frames. Since Method A's error is consistently smaller across all three frames, it provides more accurate motion estimation than Method B.

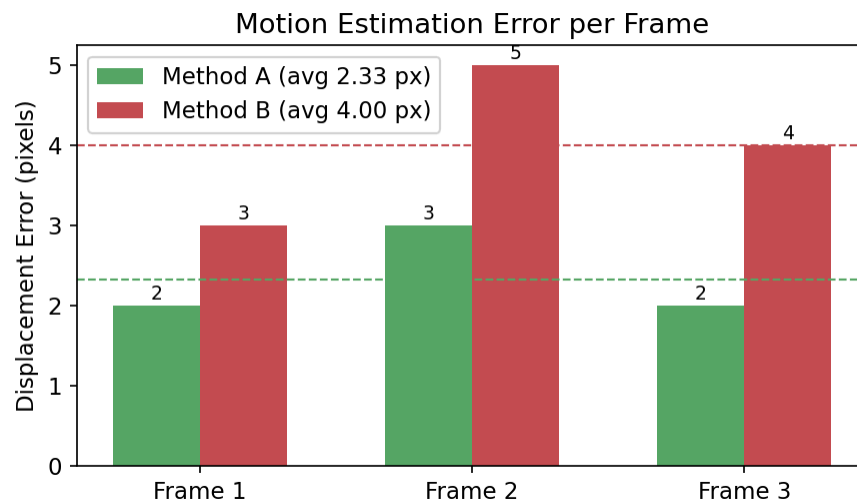


Figure 2: Frame-by-frame displacement error for Method A and Method B, with average error lines.

3. Depth Estimation Comparison

Actual depths: 2 m, 3 m, 4 m

Stereo Vision estimates: 2.0 m, 3.0 m, 4.0 m

SfM estimates: 2.5 m, 3.8 m, 5.0 m

(a) Absolute Error for Each Object

Object	Actual (m)	Stereo Estimate (m)	Stereo Abs. Error	SfM Estimate (m)	SfM Abs. Error
Object 1	2.0	2.0	$ 2.0-2.0 = 0.0$	2.5	$ 2.5-2.0 = 0.5$
Object 2	3.0	3.0	$ 3.0-3.0 = 0.0$	3.8	$ 3.8-3.0 = 0.8$
Object 3	4.0	4.0	$ 4.0-4.0 = 0.0$	5.0	$ 5.0-4.0 = 1.0$

(b) Average Absolute Error

Stereo Vision: $(0.0 + 0.0 + 0.0) / 3 = 0.0$ m

SfM: $(0.5 + 0.8 + 1.0) / 3 = 0.767$ m (approx. 0.77 m)

(c) Comparison and Justification

The Stereo Vision system has an average absolute error of 0.0 m, meaning its depth estimates exactly match the actual depths for all three objects. The SfM system has a much higher average absolute error of about 0.77 m. Since a lower absolute error indicates greater accuracy, the Stereo Vision system provides more accurate depth estimation than the Structure-from-Motion (SfM) system in this case.

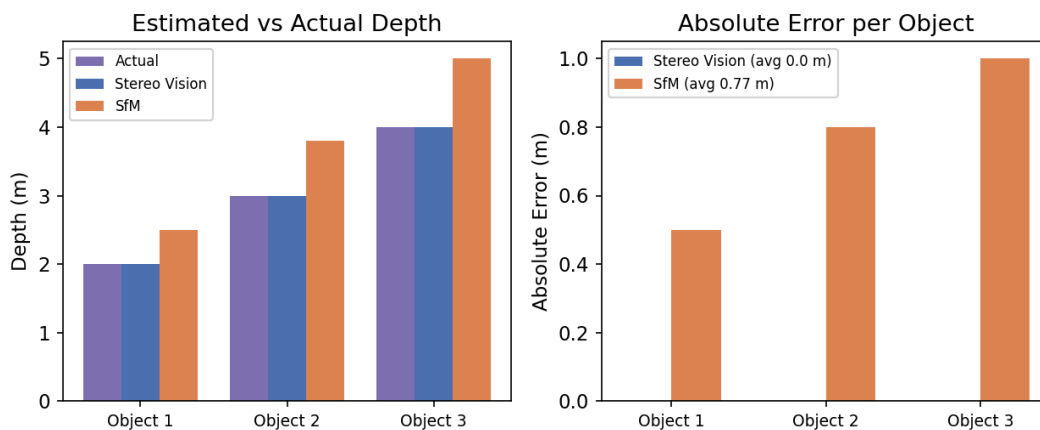


Figure 3: Estimated vs actual depth (left) and absolute error per object (right) for Stereo Vision and SfM.